

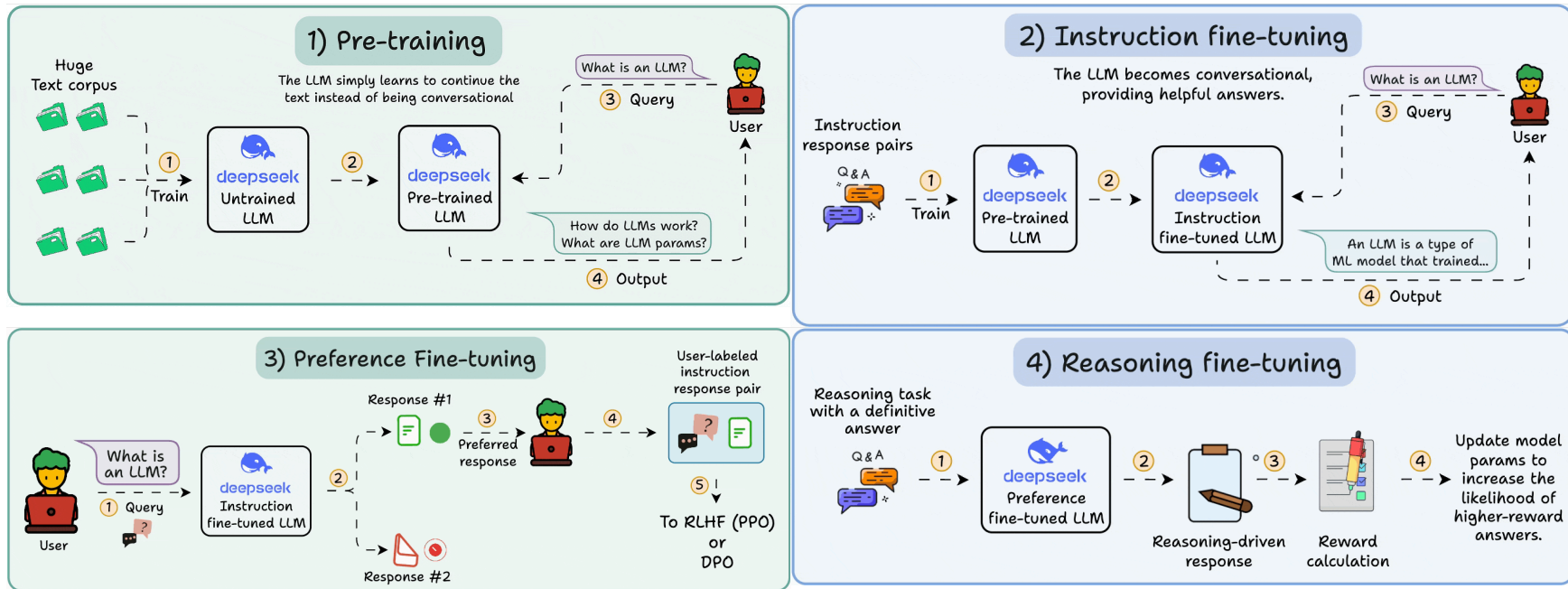


Transformer & LLM

What is Large Language Model (LLM)?

- A **LARGE** number of parameters
- **LARGE** amounts of training data

Training Stages (cite: 4 Stages of Training LLMs from Scratch, Avi Chawla)



The Origin of LLM

Attention is All You Need (slides 1, slides 2)

YONA : You Only Need Attention



1

Model Architecture

靈魂拷問：「你為什麼這樣設計？有什麼理論根據嗎？」

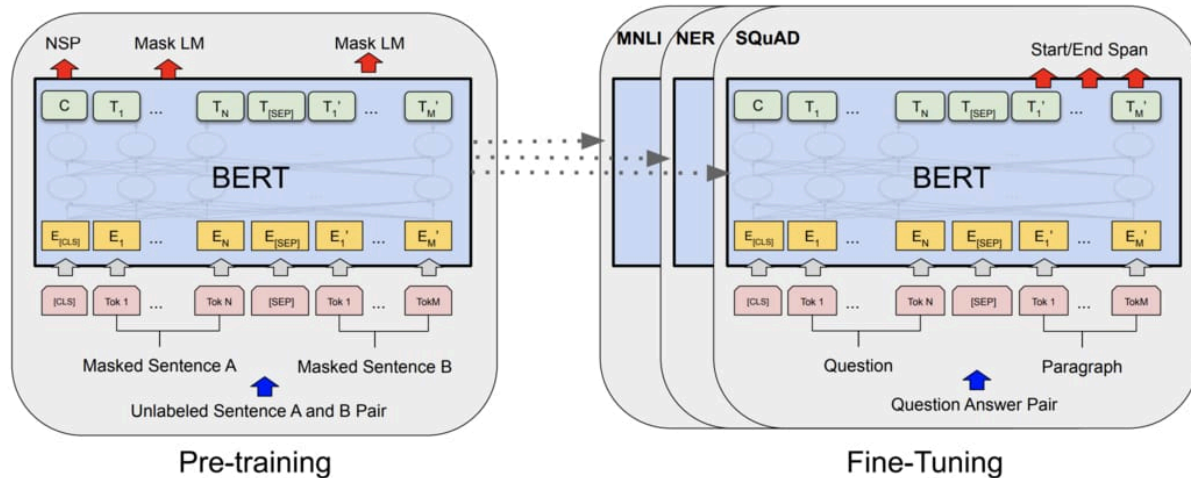
$$\left\{ \begin{array}{c} CNN \\ RNN \\ Transformer \end{array} \right\} + \left\{ \begin{array}{c} Encoder \\ Decoder \end{array} \right\} + \left\{ \begin{array}{c} Causal \\ Non-causal \end{array} \right\}$$

	Receptive Field	Memory Usage	Inductive Bias	Parallelization
CNN	受模型架構限制	$O(L)$	Y	Y
RNN	會遺失太遙遠的資訊	$O(L)$	Y	N
TNN	與模型架構無關	$O(L^2)$	N	Y

Model Architecture–BERT

抽取上下文的語意資訊

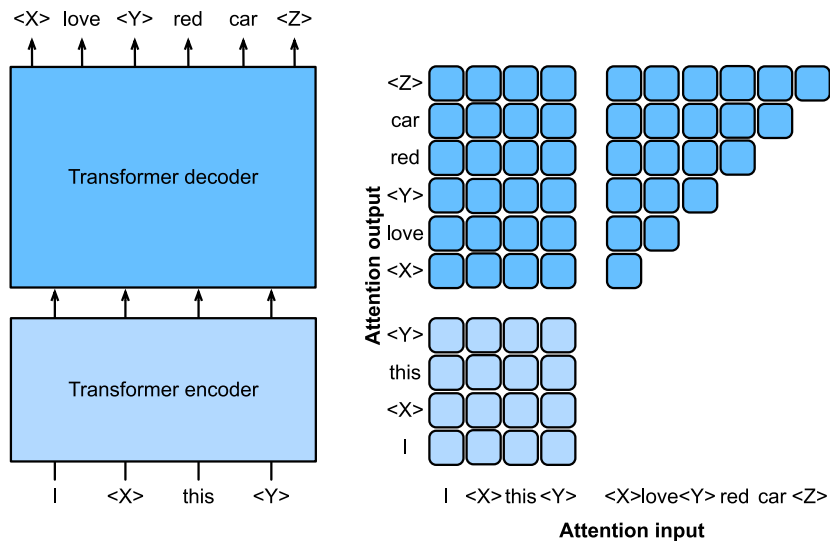
$$\left\{ \begin{array}{c} CNN \\ RNN \\ \textit{Transformer} \end{array} \right\} + \left\{ \begin{array}{c} \textit{Encoder} \\ \textit{Decoder} \end{array} \right\} + \left\{ \begin{array}{c} \textit{Causal} \\ \textit{Non-causal} \end{array} \right\}$$



Model Architecture–T5

Encoder+Decoder

$$\left\{ \begin{array}{c} CNN \\ RNN \\ \textit{Transformer} \end{array} \right\} + \left\{ \begin{array}{c} \textit{Encoder} \\ \textit{Decoder} \end{array} \right\} + \left\{ \begin{array}{c} \textit{Causal} \\ \textit{Non-causal} \end{array} \right\}$$



Pre-training Task

Original text

Thank you ~~for inviting~~ me to your party last week.

Inputs

Thank you <X> me to your party <Y> week.

Targets

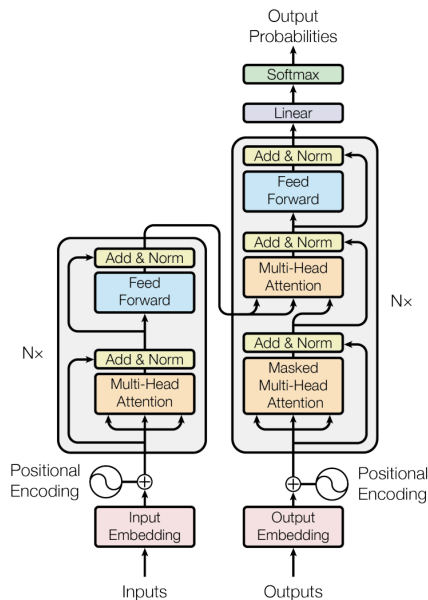
<X> for inviting <Y> last <Z>

Model Architecture–GPT

$$\left\{ \begin{array}{c} CNN \\ RNN \\ \textit{Transformer} \end{array} \right\} + \left\{ \begin{array}{c} \textit{Encoder} \\ \textit{Decoder} \end{array} \right\} + \left\{ \begin{array}{c} \textit{Causal} \\ \textit{Non-causal} \end{array} \right\}$$

BERT

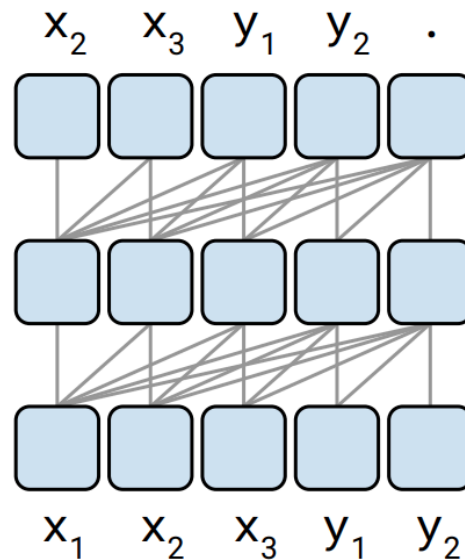
Encoder



GPT

Decoder

Pre-training Task



大力出奇蹟

- GPT: Improving Language Understanding by Generative Pre-Training
- GPT-2: Language Models are **Unsupervised Multitask Learners**
- GPT-3: Language Models are **Few-Shot Learners**

In-context Learning

Issues of LLM

Core Models & Efficiency

- Efficient Architectures
 1. Linear Attention, State Space Models
e.g. Mamba
 2. Test-Time Training
- Long-Context Processing
 1. Lost in the Middle
 2. Retrieval-Augmented Generation
 3. Length Extrapolation
 - Train Short, Test Long

Reliability & Ethics

- Hallucination
- Interpretability
- Bias & De-biasing
 1. Training Free Guidance e.g. Linear Alignment
 2. Unlearning

Applications & Interaction

- Agent AI e.g. ReAct, ReWOO, AutoGPT
- Domain Adaptation
- Emergent Abilities
 1. Thoughts e.g. CoT, GoT, XoT, FoT
 2. Reasoning

